MASTER PROJECT

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US–Mexico migration corridors

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Abstract

1. Introduction

International remittances are one of the main private channels through which local economic shocks can cross national borders. Mexico is a central setting in which to study this mechanism. Banco de México records US$64.7 billion in family remittance receipts in 2024, while World Bank estimates place Mexico among the largest remittance recipients in the world, second only to India (**banxico_remittances_ca11**; World Bank, 2024). The U.S.–Mexico corridor is especially important because Mexican households receive transfers from migrants whose earnings are tied to local labor markets across the United States. This paper asks whether a negative local shock in the United States is transmitted to Mexican municipalities through remittances. Do migrants smooth transfers after being hit by a shock, or do origin households bear part of the adjustment through lower remittance inflows?

The answer matters because remittances are not only an aggregate balance-of-payments flow, they are also part of household resources. Our data do not directly observe Mexican household income or consumption. Even so, changes in remittance inflows are economically meaningful for household welfare. Prior work shows that remittances are an important source of household resources in Mexico, affecting consumption, poverty, and inequality (**adamsetal2005**), food security (**MORARIVERA2021105349**), and schooling and child labor decisions (**ALCARAZ2012156**). We therefore interpret changes in municipal remittance inflows as reduced-form evidence on a channel with direct relevance for household income and consumption, while not claiming to estimate those household outcomes themselves.

The main empirical difficulty is that most large shocks are not cleanly assigned to one side of the migration corridor. Caballero et al. (2023) show that Mexican households connected to U.S. labor markets more severely affected by the Great Recession were less likely to receive remittances, providing important evidence that migrant networks transmit U.S. shocks to Mexico. Yet the Great Recession simultaneously affected the United States and Mexico, making it difficult to isolate the remittance channel from other aggregate forces. Their analysis also focuses on the extensive margin of remittance receipt. Quantifying how much bilateral remittance flows adjust on the intensive margin is necessary to measure the income loss experienced by origin households.

We address these challenges by exploiting Hurricane Ian, which made landfall in southwestern Florida on September 28, 2022. Ian caused an estimated US$109.5 billion in damages in Florida and was the costliest hurricane in the state's history (Bucci et al., 2026). The shock was severe and spatially concentrated on the sender side of the corridor: it disrupted economic activity in Florida but did not directly affect Mexican municipalities. Natural disasters in the United States can reduce local productivity, labor demand, housing wealth, and household resources (Boustan et al., 2020). Hurricane Ian therefore provides a localized negative shock to the capacity of Florida-based migrants to remit, while leaving recipient municipalities in Mexico exposed only through their pre-existing migration links to Florida.

We base our argument on the following idea. A shock to Florida migrants can directly reduce remittances from Florida to Mexican municipalities. However, many origin municipalities receive remittances from migrants in several U.S. states. If migrants care about total resources available to the origin household, then a fall in Florida remittances can raise the marginal value of transfers from migrants living elsewhere. The same shock can therefore generate two effects at once: a direct decline in Florida-to-Mexico remittances and a compensatory response from other U.S. states connected to the same municipalities.

The first part of the paper formalizes this idea with a structural remittance model. The model follows the insight in **albert_monras_2022<empty citation>** that migrants allocate resources across where they live and where their origin households are located. In our setting, this idea is applied to remittance choices rather than to location choices. Migrants choose between local consumption in the United States and resources available to the origin household in Mexico. They differ in income and in the strength of their altruistic or obligation motive, and remitting involves both a fixed cost and a variable cost. In the spirit of models with heterogeneous agents, fixed costs, and variable costs (Melitz, 2003), the fixed cost generates an extensive margin of remitting, while the variable cost governs the intensity of transfers.

The key extension is that origin-household resources include local non-remittance income plus remittances received from migrants in other U.S. states. This makes the model directly useful for interpreting the empirical patterns. A negative shock to Florida migrants reduces their optimal remittances. At the same time, for households linked to Florida, the fall in Florida transfers lowers perceived origin resources for migrants in other states, raising the marginal utility of additional remittances under an insurance motive. The model therefore provides a unified way to interpret both the direct exposure effect and the compensatory network response observed in the data.

The second part of the paper brings the model to the data by constructing a bilateral panel of remittance flows from U.S. states to Mexican municipalities. Because such flows are not directly observed, we combine two sources from Banco de México with MCAS administrative records. Banco de México reports remittance inflows by U.S. state of origin and by Mexican municipality of receipt (Banco de México, CE166, 2026; Banco de México, CE168, 2026). MCAS records identify the Mexican municipality of origin and U.S. state of residence of Mexican consular-card holders, providing a proxy for the geography of migrant networks. We use these records to construct state-to-municipality migration shares and allocate observed state-level remittance outflows across Mexican municipalities.

Two validation exercises support this construction. First, the migration network data are highly persistent: the distribution of Mexican migrants across U.S. states and Mexican municipalities has year-to-year correlations above 0.90 from 2012 onward. Second, when we use these migration shares to allocate state-level remittance outflows to Mexican municipalities, the resulting municipal aggregates correlate between 0.75 and 0.85 with official municipal remittance receipts. These facts suggest that the estimated corridors capture meaningful variation in the geography of remittances.

The third part of the paper studies the shock in three steps. First, we examine remittance outflows from Florida after Hurricane Ian and observe a decrease in the data, consistent with a sender-side shock reducing migrants' capacity to remit. Second, we exploit variation in Mexican municipalities' pre-existing exposure to Florida to estimate whether Ian affected municipal remittance inflows differently depending on their exposure to the shocked corridor. In the period immediately following Ian, a 1% increase in a municipality's pre-existing exposure to Florida is associated with a 0.16% decrease in remittances received. This result is consistent with a sender-side income shock being transmitted through the remittance corridor to origin municipalities.

Finally, we examine migration-network spillovers. Among municipalities jointly exposed to Florida and to other large sender states, remittances from California and Texas rise relative to

Florida in the first post-Ian quarters. The diagnostic plots show short-run increases of roughly 20–40 percent for these alternative corridors before the response fades and eventually reverses. This pattern is consistent with compensatory transfers from migrants in other states.

Related Literature. The paper is closest to work that studies remittances as a form of household insurance (Bettin et al., 2023; Yang & Choi, 2007). Most existing evidence studies shocks in receiving countries, where migrants abroad may increase transfers to insure households at origin. We instead study a sender-side shock, where the capacity to remit changes in the destination labor market while origin municipalities are not directly affected. This distinction is important because the same remittance channel that insures households against origin shocks may transmit destination shocks back to those households.

The paper also relates to research on migrant networks and the international transmission of local shocks (Caballero et al., 2023; Munshi, 2003). The closest paper is Caballero et al. (2023), which shows that Mexican households linked to U.S. labor markets hit by the Great Recession became less likely to receive remittances. We differ by studying a localized natural disaster, constructing bilateral state-to-municipality remittance corridors, and focusing on intensive-margin flows and cross-corridor responses.

The structural framework connects to research that models migrants as allocating resources across origin and destination locations (**albert_monras_2022**). The difference is that our model uses this logic to study remittance behavior after a destination-side shock. It also connects to work on interference and spillovers in causal inference (Aronow & Samii, 2017; Butts, 2023; Hudgens & Halloran, 2008). In our setting, interference is not only a threat to identification. It is part of the economic mechanism, because migrants linked to the same origin municipality may jointly insure households against shocks that affect only one destination.

The remainder of the paper is organized as follows. Chapter 2 develops the remittance model, describes Hurricane Ian as the natural experiment, and presents the empirical strategy. Chapter 3 describes the construction and validation of the bilateral remittance panel. Chapter 4 reports the estimates for direct effects, exposure effects, and spillovers. Chapter 5 concludes.

2. Theoretical framework and empirical strategy

This chapter outlines the theoretical and empirical framework use for our analysis. We begin by adapting a heterogeneous-firm trade model to the context of individual remittance behavior, enstablishing the theoretical foundations of how a migrant's transfer decisions can be influenced by economic shocks. Next, we introduce Hurricane Ian, which hit Florida in 2022, as a natural experiment that acts as an exogenous shock to migrant income. Finally, we present our estimation strategy (explain the strategy here) and discuss how we examine potential spillover effects, which can bias our estimates if not properly accounted for.

2.1. Remittance model

To analyze the impact of shocks on remittance flows, we adapt the logic of heterogeneous-firm trade models to the context of remittances. Specifically, we consider a model where migrants differ in income and altruism intensity, and where remitting requires paying both a fixed access cost and a variable cost proportional to the amount remitted. Formally, consider migrant i from origin municipality m in year t. The migrant faces a utility maximization problem under a CES aggregator over local consumption and total resources available to the origin household:

$$U_{imt} = \left[\alpha \left(c^L_{imt}\right)^{\frac{\sigma-1}{\sigma}} + (1-\alpha)\left(\phi_{im}\left(R_{imt} + y^O_{mt}\right)\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{2.1}$$

where $\phi_{im} > 0$ denotes the altruism or obligation intensity parameter, analogous to the productivity parameter in Melitz (2003). This stricly positive nature of this parameter ensures that sending remittances always increases the migrant's utility. The variable $c^L_{imt} \geq 0$ represents the migrant's consumption in the destination country, $R_{imt} \geq 0$ is the amount sent in remittances, and $y^O_{mt} \geq 0$ captures the origin hosehold's income as observed by the migrant. The weight of local consumption is given by $\alpha \in (0, 1)$, and the elasticity of substitution between local consumption and origin-household resources is given by σ. Notably, origin-household total resources are defined as $R_{imt} + y^O_{mt}$, which we treat as a single composite good. The presence of y^O_{mt} introduces a role for origin-country economic conditions: negative income shocks at origin (i.e. lower y^O_{mt}) increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007).

The migrant's budget constraint is given by:

$$c_{imt}^{L} + f_{mt}\mathbb{1}\{R_{imt} > 0\} + (1+\tau_{mt})R_{imt} \leq w_{imt} \tag{2.2}$$

where $w_{imt} \geq 0$ is the migrant's wage, $f_{mt} \geq 0$ is a fixed remitting cost capturing the per-transaction expense of accessing a remittance channel, analogous to the fixed export cost in Melitz (2003), and $\tau_{mt} > -1$ is the variable cost which scales with the amount remitted, analogous to the iceberg cost structure in Melitz (2003). Notice that, even though τ_{mt} is typically positive to reflect transaction fees or exchange rate spreads, we allow for the possibility of negative variable costs, which could arise from the presence of remittance subsidies or favorable exchange rate regimes.

Assuming that the migrant maximizes utility subject to the budget constraint and chooses to remit a positive amount ($R_{imt} > 0$), solving the Lagrangian yields the optimal remittance function:

$$R_{imt}^{*} = \frac{\left(w_{imt} - f_{mt} + (1+\tau_{mt})y_{mt}^{O}\right) A_{imt} - y_{mt}^{O}}{1 + (1+\tau_{mt})A_{imt}} \tag{2.3}$$

where we define the parameter A_{imt} as:

$$A_{imt} \equiv \left(\frac{(1-\alpha)\phi_{im}}{\alpha(1+\tau_{mt})}\right)^{\sigma} \tag{2.4}$$

The term A_{imt} represents the optimal ratio of origin-household resources to own local consumption, adjusted for the variable cost of remitting. It comprises the migrant's altruism intensity (ϕ_{im}), the utility weight of own consumption (α), the elasticity of substitution (σ), and the variable cost of remitting (τ_{mt}).

To understand the effect of destination-country economic shocks on remittance flows, we analyze changes in the migrant's wage (w_{imt}). The partial derivative of optimal remittances with respect to wage is strictly positive:

$$\frac{\partial R_{imt}^{*}}{\partial w_{imt}} = \frac{A_{imt}}{1 + (1+\tau_{mt})A_{imt}} \tag{2.5}$$

Therefore, through an income effect, the model predicts that a positive shock to the migrant's wage will increase remittances, while a negative shock will decrease them. The fixed cost f_{mt} enstablishes an entry threshold: if the migrant's wage is not sufficiently high relative to the fixed cost, they will not send remittances at all. Additionally, the variable cost τ_{mt} reduces the marginal benefit of each dollar sent, dampening the sensitity of remittances to wage flactuations.

Ultimately, this model provides a theoretical framework to understand how shocks that affect migrants' income can influence remittance flows, which is the main focus of our empirical analysis. However, a negative shock to migrant's wage does not directly imply a significant decrease in remittances, as the response depends on the shock's magnitude, the migrant's altruism intensity, and the cost structure of remitting.

2.2. The natural experiment: Hurricane Ian (2022)

The shock selected for this analysis is Hurricane Ian, which struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph) (Bucci et al., 2026). Ian caused an estimated US$ 112.9 billion in total damage in the United States, of which US$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida's history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean. Indeed, remittance senders in Florida were directly exposed to a severe and sudden wealth shock (reference or data?), while remittance receivers in Mexican municipalities were entirely unaffected. This allows us to better isolate the transmission of the shock through the remittance channel.

2.3. Estimation technique

To estimate the impact of the Hurricane Ian shock on Florida remittance outflows, we employ three complementary panel-data estimators: Two-Way Fixed Effects (TWFE), Synthetic Control (SC), and Synthetic Difference-in-Differences (SDID). In all specifications, Florida is treated as the exposed unit, while the remaining U.S. states constitute the donor pool. The outcome variable is the logarithm of remittance outflows, and the sample period spans from 2013Q1 to 2024Q4. The treatment period begins in 2022Q4, corresponding to the landfall of Hurricane Ian. To account for the potentially temporary nature of disaster-induced remittance responses, treatment effects are estimated over several post-treatment horizons, including the full post-treatment sample as well as restricted windows covering the first four and six post-treatment quarters.

The baseline econometric specification is a standard Two-Way Fixed Effects (TWFE) model of the form:

$$Y_{it} = \alpha_i + \gamma_t + \beta(Florida_i \times Post_t) + \varepsilon_{it}$$

where Y_{it} denotes the logarithm of remittance outflows for state i in quarter t, α_i represents state fixed effects, and γ_t denotes time fixed effects. The interaction term $Florida_i \times Post_t$ captures the average treatment effect of Hurricane Ian on Florida relative to the control states. Standard errors are clustered at the state level. Additional TWFE specifications incorporate state-specific linear trends and seasonal fixed effects to account for differential long-run trajectories and recurrent seasonal patterns in remittance flows. The identifying assumption underlying TWFE is the parallel trends assumption, which is evaluated using an event-study design estimating dynamic leads and lags of treatment relative to the quarter immediately preceding the hurricane.

To complement the TWFE analysis, we estimate a Synthetic Control (SC) model following the framework of (**abadie2015synthetic**) The SC estimator constructs a weighted average of untreated states that minimizes the pre-treatment discrepancy between Florida and its synthetic counterpart. Formally, the estimator selects weights w_j such that:

$$\sum_{j=1}^{J} w_j Y_{jt}$$

closely reproduces Florida's pre-treatment outcome trajectory, subject to $w_j \geq 0$ and $\sum_j w_j = 1$. The estimated treatment effect is then computed as the post-treatment gap between observed Florida remittances and the synthetic Florida series. Inference is conducted through placebo reassignments, where each control state is iteratively treated as if exposed to Hurricane Ian, generating a placebo distribution of treatment effects. We additionally compute Root Mean Squared Prediction Error (RMSPE) ratios comparing pre- and post-treatment fit to evaluate whether Florida exhibits unusually large post-treatment deviations relative to placebo units.

Finally, we estimate a Synthetic Difference-in-Differences (SDID) model, following the methodology proposed by Arkhangelsky et al. (2021). SDID combines the weighting structure of Synthetic Control with the fixed-effects adjustment of Difference-in-Differences, thereby improving robustness to imperfect pre-treatment fit and treatment effect heterogeneity. The estimator jointly constructs unit weights and time weights to balance pre-treatment outcomes between treated and control units before applying a Difference-in-Differences correction. The SDID estimator can be expressed as:

$$\hat{\tau}^{SDID} = \left(\bar{Y}^{post}_{treated} - \sum_i \omega_i \bar{Y}^{post}_i\right) - \left(\sum_t \lambda_t Y^{pre}_{treated,t} - \sum_i \omega_i \sum_t \lambda_t Y^{pre}_{it}\right)$$

where ω_i denote unit weights and λ_t denote time weights estimated from the pre-treatment period. Confidence intervals for SDID estimates are computed using placebo-based variance estimators. Together, the three estimators provide a comprehensive framework for evaluating the robustness of the estimated hurricane effects under different identifying assumptions and counterfactual constructions.

2.4. Examining spillover effects

A fundamental requirement for standard causal inference models is the Stable Unit Treatment Value Assumption (SUTVA), formalized by Rubin (1980). SUTVA assumes that the potential outcomes of any unit depend exclusively on its own treatment assignment, excluding the possibility of any indirect spillover between units.

In our context, U.S. states not directly exposed to a local shock might still be affected if control units are exposed to the treatment through indirect channels, leading to biased estimates (Clarke, 2017). For instance, given that California and Texas all host a big proportion of Mexican migrants, we hypothesize later in the paper that these two states might have played a role in offsetting the negative Florida shock. If a Mexican municipality highly dependent on Florida remittances experiences an income shock due to Hurricane Ian, connected migrants residing in Texas might respond by increasing their transfers to compensate for the drop. If this was the case, **the estimated Florida-exposure effect on total remittances would capture the net effect of the shock after within-network adjustment, rather than the direct fall in Florida-origin remittance**.

Therefore, in later sections we will try to argue that our estimates are subject to general equilibrium effects. We base our analysis on the exposure-mapping logic of (Aronow & Samii, 2017) to define indirect exposure through alternative migrant networks. Rather than assuming that a municipality's remittances respond only to Florida exposure, we allow the response to vary with exposure to other major U.S. destinations, especially Texas and California.

3. Data and bilateral remittance flow estimation

This chapter outlines the data sources and the methodology we construct for estimating bilateral remittance flows between U.S. states and Mexican municipalities, which is not publicly available. First, we describe the datasets from Banco de México and Matricula Consular de Alta Seguridad (MCAS), which provide the remittance and migration data, respectively. Then, we compute a migration-share weighting matrix, which we use to distribute U.S. state-level remittance outflows to Mexican municipalities. Finally, we explore the underlying assumptions of our estimation approach, testing the persistence of migration networks over time and validating our estimates by comparing them with official data available at a more aggregate level.

3.1. Data sources

The source of remittance data is Banco de México (Banxico), which provides two complementary datasets through its Sistema de Información Económica (SIE). The first, series CE168, records remittance inflows from each US state to Mexico in millions of current US dollars at quarterly frequency, covering the period from Q1 2013 to Q1 2026 (Banco de México, CE168, 2026). This series serves as the US state ourflows (r_s) for our bilateral flow estimation. The second, series CE166, records remittance inflows received by each Mexican municipality, also in millions of current US dollars at quarterly frequency and over the same sample period (Banco de México, CE166, 2026). This series acts as abenchmark to validate the accuracy of our estimates at an aggregate level.

To bridge these datasets, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular identification cards issued to Mexican nationals residing in the United States each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.2. Bilateral remittance estimation

To estimate bilaterance remittance flows from U.S. states to Mexican municipalities, we construct a migration-based weighting matrix using annual MCAS from 2010 to 2024. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistance of migration patterns over time. To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of these matrices across all years. We then apply this averaged weighting matrix to the state-level remittance outflows from Banxico's series CE168 to derive the U.S. to Mexican municipality bilateral flows.

Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s. For each U.S. state, we calculate the average migration share, w_{ms}, which represents the proportion of that state s's Mexican population originating from the m municipality, as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances from that state across all municipalities M:

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{m=1}^{M} MCAS_{ms}} \tag{3.1}$$

By applying the weights computed from equation (3.1) to the official aggregate state-level outflows from Banxico (r_s), we derive the estimated bilateral remittance flow ($\hat{r}_{ms}$) from state s to municipality m:

$$\hat{r}_{ms} = r_s \cdot w_{ms} \tag{3.2}$$

This migration share estimation approach yields the final estimate of our outcome variable.

3.3. Underlying assumptions and validation exercises

The validity of this estimation relies on two primary identification assumptions. First, we assume the temporal persistence of migration flows, implying that the distribution of migrants recorded in the 2010–2024 MCAS data is representative of the active remitting population.

Second, we assume that, within a given U.S. state, the average amount remitted does not vary systematically by the migrant's municipality of origin. While this assumption ignores micro-level heterogeneity, it provides a baseline estimate of the bilateral remittance flows that can be validated against the aggregate data provided by Banxico' series CE166.

3.3.1. *Persistence of migration networks*

The persistence assumption of migration networks is critical to our estimation strategy, as it justifies the use of MCAS data to construct the migration-based weighting matrix in order to allocate remittance flows. The stability of migration patterns over time is well-documented in the literature, with studies showing that enstablished migration networks tend to dictate future migration patterns, creating persistent flows between specific origin and destination pairs (Card, 2001; Munshi, 2003).

To verify this assumption within our specific sample period, we conduct a validation exercise using the MCAS data by computing the Pearsoncorrelation of our constructed weighting matrices across years. As shown in Figure 3.1, when taking any year from 2012 onward as the base year, the correlation with subsequent years is consistently above 0.90. While the 2010 and 2011 matrices exhibit strong correlation between them, their correlation with later years is smaller, specifically between 0.75 and 0.85. This may be due to structural adjustment in migration patterns following the Great Recession.

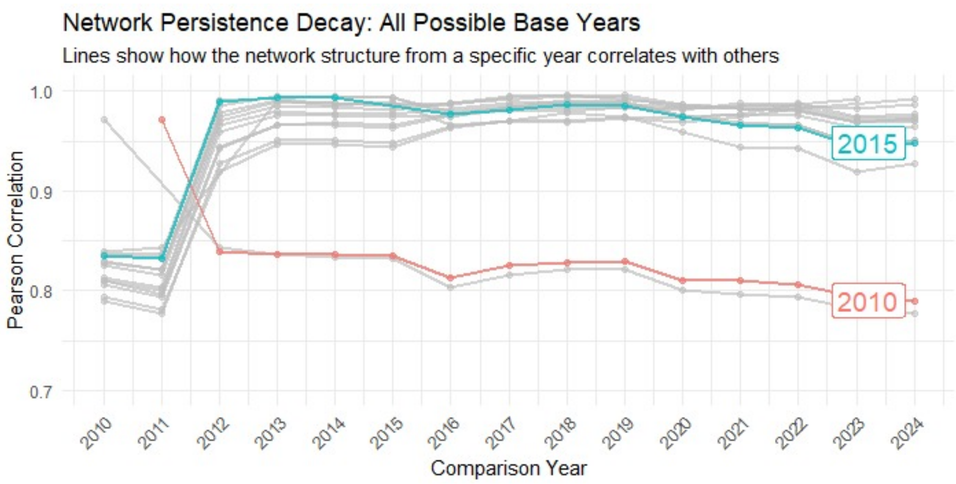

Figure 3.1: Correlation of MCAS-based weighting matrices across years. Each pointrepresents the correlation coefficient between the weighting matrix of the base year and the weighting matrix of the comparison year.

Source: Authors' own elaboration based on MCAS data.

This high degree of correlation suggests that the spatial distribution of Mexican migrants across U.S. states has remained highly stable over time, lending credibility to our use of an averaged weighting matrix for estimating bilateral remittance flows.

3.3.2. *Validation of the estimated remittance flows*

Our methodology to estimate bilateral remittance flows implicitly assumes a homogeneous send behavior among migants withing a given U.S. destination. That is, migrants in U.S. state s send on average the same amount of remittances, regardless of their specific Mexican municipality of origin m. To evaluate the empirical validity of this assumption, we aggregate our estimated bilateral municipal inflows and compare them against the observed municipal receipts reported in Banco de México's series CE166.

Figure 3.2 plots the Pearson correlation between our estimates aggregated at the municipal level and the official inflows for each year. The results report correlation coefficients consistently between 0.75 and 0.85,suggesting that our estimation procedure successfully captures a significant portion of the variation in remittance inflows across Mexican municipalities. While this result validates our baseline approach, we acknowledge that it does not capture all the heterogeneity in remittance flows, which is expected given the simplifying assumptions we made.

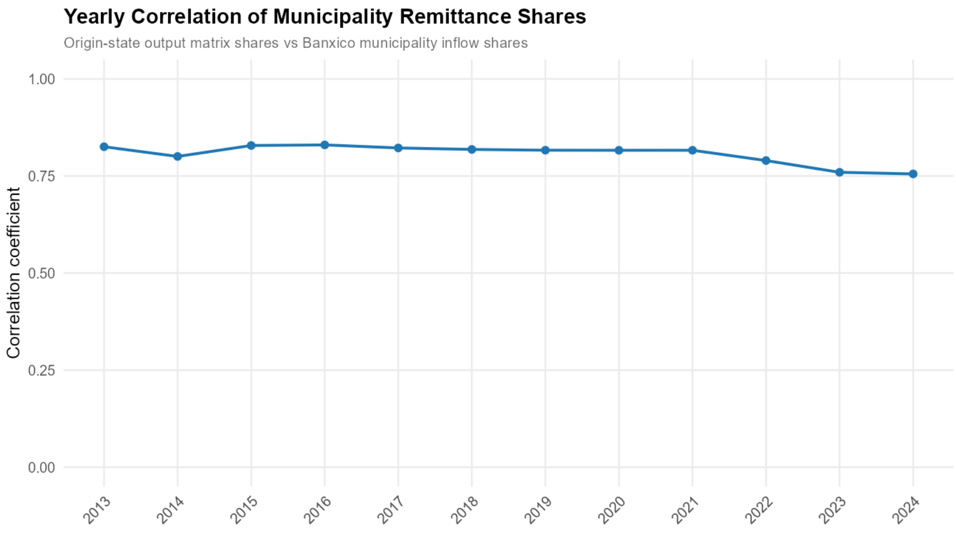

Figure 3.2: Correlation of estimated municipal remittance inflows with official data. Each point represents the Pearson correlation coefficient between the estimated municipal inflows (aggregated from our bilateral estimates) and the actual municipal inflows recorded in Banxico's series CE166 for a given year.

Source: Authors' own elaboration.

4. Results

4.1. Preliminary statistics

check parallel trends

statistics before (e.g. average over the past year), 1, 2, 3, and 4 quarters after the shock. I would do this with growth rates even if the estimation is in levels because quarterly growth rates are more interpretable and take into account the trend. Note that there is also seasonality to take into account.

4.2. DiD results

4.3. Arguing for Potential Spillover Effects

As argued above, given that the biggest hosts of Mexican migrants are Texas and California, it is likely that those municipalities exposed to the Florida shock might also be connected to a certain extent to California and Texas. Plot 4.1 helps us visualize that there exists indeed some migrant networks spanning across Florida and California and Florida and Texas.

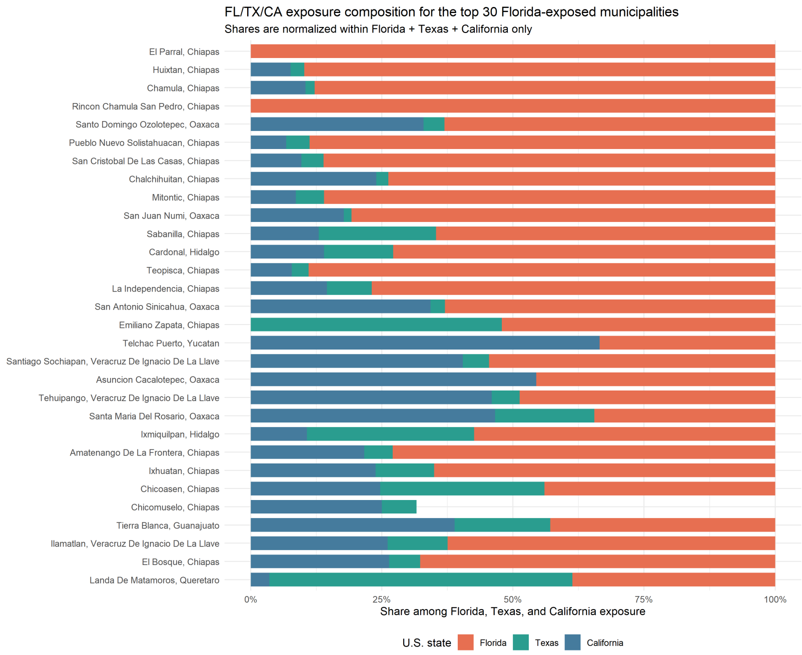

Figure 4.1: For the Top 30 Florida exposed municipalities, the plot shows the composition of migrants spread across the three US states.

Figure 4.2 suggests that there has been a dip in all 3 states in 2022Q4, after the Hurricane Ian hit Florida. However, we are interested in examining **how the inflow of remittances from Texas and California evolved around the shock relative to the evolution of Florida inflows**, specifically for the municipalities that are jointly highly exposed to Florida and California or Florida and Texas, provided there exist **parallel trends in the remittance patterns between these two groups (WE NEED PPML MAYBE!!!)**.

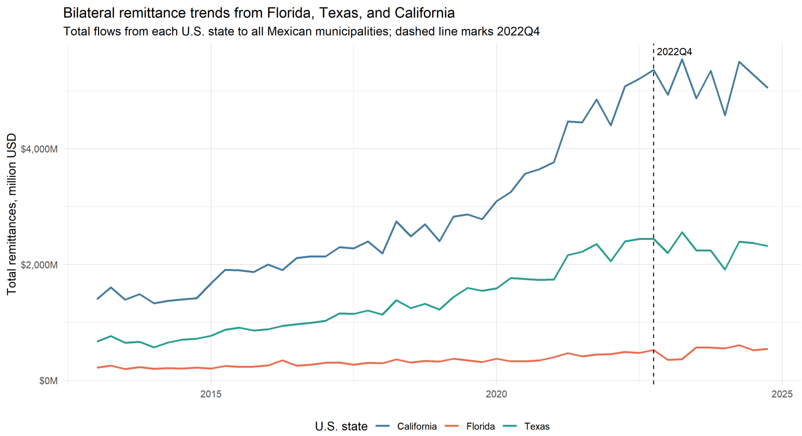

Figure 4.2: Evolution of Remittances Inflows from FL, TX and CA to all Mexican municipalities

Let's define a measure of joint exposure FL-TX of Mexican municipality m as the geometric mean of its exposure to Florida and its exposure to Texas, as follows:

$$JointExposure_m^{FL,TX} = sqrt(Exposure_m^{FL} \times Exposure_m^{TX})$$

Similarly, we can also define a measure of joint exposure FL-CA. As an example, 4.1 ranks the municipalities by highest joint exposure to both Florida and Texas.

We now define as "High joint exposure" as the top quartile municipalities with the highest joint exposure and "Low joint exposure" as the bottom quartile municipalities with the lowest joint exposure and estimate two log remittances regressions under the following specifications:

$$\log(1 + R_{mst}) = \alpha_t + \gamma_{ms} + \beta_{TX}(Texas_s \times Post_t) + \varepsilon_{mst}$$
$$\log(1 + R_{mst}) = \alpha_t + \gamma_{ms} + \beta_{CA}(California_s \times Post_t) + \varepsilon_{mst}$$

Note that in the TX-FL regression, the sample is restricted to the highly exposed municipalities (As defined above). Similarly, in the CA-FL regression, the sample is also restricted to the cor-

Table 4.1: Top 10 Joint Exposure FL-TX

Rank	State	Municipality	GeoMean	GeoMean %
1	Chiapas	Emiliano Zapata	0.3373	33.7261
2	Queretaro	Landa De Matamoros	0.3262	32.6221
3	Hidalgo	Pisaflores	0.3157	31.5685
4	Veracruz	Ignacio De La Llave	0.2703	27.0324
5	Michoacan De Ocampo	San Lucas	0.2673	26.7327
6	Chiapas	La Libertad	0.2539	25.3916
7	Veracruz	Tihuatlan	0.2532	25.3234
8	Chiapas	Chicoasen	0.2391	23.9073
9	Chiapas	Sabanilla	0.2371	23.7081
10	Chiapas	Rayon	0.2348	23.4797

responding high jointly exposed municipalities.

We include α_t as the quarter FE, γ_{ms} as the corridor MX municipality - US State FE and $Post_t$ is defined such that $Post_t = 1$ for quarters from 2022Q4 onward, and 0 otherwise. Since Florida is the baseline category, the coefficient of interest β_{TX} (or β_{CA}) compares the post-2022Q4 change in Texas remittances to the post-2022Q4 change in Florida remittances (among our subsamples of interest). To ensure robustness of results, we also clustered at corridor and quarter level.

In other words, the coefficients β_{TX} and β_{CA} capture respectively:

$$\begin{aligned} \beta_{TX} &= [\Delta \log R_{m,Texas}] - [\Delta \log R_{m,Florida}] \\ \beta_{CA} &= [\Delta \log R_{m,California}] - [\Delta \log R_{m,Florida}] \end{aligned}$$

The following figure 4.3 shows that in relative terms, the inflows of remittances from Texas and California to high joint exposure municipalities have experienced a significant increase in the two quarters posterior to the Hurricane shock, followed by an immediate decline. This could suggest a **short-term change of remittance patterns of these subsamples** right after the shock: municipalities that are strongly related to both Florida and the California or Texas might have been **relatively less affected by the Florida shock since they also received relatively more inflows of remittances from the other state that could potentially offset the decrease in inflows from FL.**

In other words, it could also suggest that the REVISE: the estimates we found in the earlier sections on the drop in remittances received by Florida households are actually attenuated by these potential spillover effects.

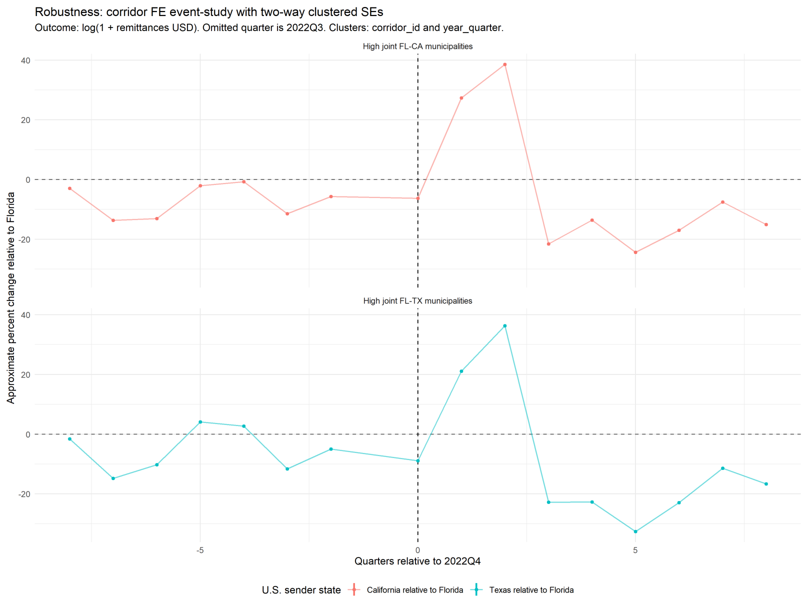

Figure 4.3: Change in log remittances inflows from TX and CA rel. to FL to high joint exposure municipalities

Despite the increase of remittances to high exposure municipalities from Texas and California relative to Florida, in order to support our intuition and confirm the existence of spillovers, we would need to check a for parallel trends between the high and low exposure municipalities and their inflows of remittances from Texas/California. In other words, we do not want to **confound the compensating effect that has risen due to the Hurrican shock with a general upward trend in remittances received in Texas relative to Florida**, for instance.

here, we should actually test for parallel trends. control: low exposure municipalities TX-FL, treatment: high exposure municipalities TX-FL. We want to check if the trends in remittances received from Texas relative to Florida were parallel between the high and low exposure municipalities before the shock. If they were parallel, then we can be more confident that the divergence we see after the shock is indeed due to the shock and not due to some pre-existing trend.

5. Conclusion

Bibliography

Aronow, P. M., & Samii, C. (2017). Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, *11*(4), 1912–1947. https://doi.org/10.1214/16-aoas1005

Banco de México, CE166. (2026). *Ingresos por remesas, distribución por municipio (CE166)* [Sistema de Información Económica (SIE), Banco de México]. Retrieved May 7, 2026, from https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE166&locale=es

Banco de México, CE168. (2026). *Estado de origen de los ingresos por remesas provenientes de Estados Unidos (CE168)* [Sistema de Información Económica (SIE), Banco de México. Trimestral, millones de dólares, flujos. Periodo: Ene-Mar 2013 – Jul-Sep 2024]. Retrieved May 7, 2026, from https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE168&locale=es

Bettin, G., Jallow, A., & Zazzaro, A. (2023). *How do monthly remittances respond to natural disasters in migrants' home countries?* (Mo.Fi.R. Working Papers No. 179). Money, Finance Research group (Mo.Fi.R.) - Univ. Politecnica Marche - Dept. Economic, and Social Sciences. https://doi.org/None

Boustan, L. P., Kahn, M. E., Rhode, P. W., & Yanguas, M. L. (2020). The effect of natural disasters on economic activity in us counties: A century of data. *Journal of Urban Economics*, *118*, 103257. https://doi.org/https://doi.org/10.1016/j.jue.2020.103257

Bucci, L., Alaka, L., Hagen, A., Delgado, S., & Beven, J. (2026). *Tropical cyclone report: Hurricane Ian (AL092022), 23–30 September 2022* (tech. rep.). National Hurricane Center, National Oceanic and Atmospheric Administration (NOAA). Retrieved May 7, 2026, from https://www.nhc.noaa.gov/data/tcr/AL092022_Ian.pdf

Butts, K. (2023). Difference-in-differences estimation with spatial spillovers. https://arxiv.org/abs/2105.03737

Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2018). Measuring geographic migration patterns using matrículas consulares. *Demography*, *55*(3), 1119–1145. https://doi.org/10.1007/s13524-018-0675-6

Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2023). The international transmission of local economic shocks through migrant networks. *Journal of International Economics*, *145*, 103832. https://doi.org/https://doi.org/10.1016/j.jinteco.2023.103832

Card, D. (2001). Immigrant inflows, native outflows, and the local labor market impacts of higher immigration. *Journal of Labor Economics*, *19*(1), 22–64.

Clarke, D. (2017). *Estimating difference-in-differences in the presence of spillovers* (MPRA Paper No. 81604). Munich Personal RePEc Archive (MPRA). https://www.damianclarke.net/research/papers/Clarke_Spillovers.pdf

Hudgens, M. G., & Halloran, M. E. (2008). Toward causal inference with interference. *Journal of the American Statistical Association*, *103*(482), 832–842. http://www.jstor.org/stable/27640105

Instituto de los Mexicanos en el Exterior. (2026). Estadísticas: Información estadística de la población mexicana en el exterior [Accessed: 2026-04-01]. *Secretaría de Relaciones Exteriores (SRE)*.

Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, *71*(6), 1695–1725.

Munshi, K. (2003). Networks in the modern economy: Mexican migrants in the U. S. labor market. *The Quarterly Journal of Economics*, *118*(2), 549–599.

Rubin, D. B. (1980). Randomization analysis of experimental data: The fisher randomization test comment. *Journal of the American Statistical Association*, *75*(371), 591–593. http://www.jstor.org/stable/2287653

World Bank. (2024). Personal remittances, received (current US$) [World Bank Open Data. Accessed: 2026-05-07]. https://data.worldbank.org/indicator/BX.TRF.PWKR.CD.DT

Yang, D., & Choi, H. (2007). Are remittances insurance? evidence from rainfall shocks in the Philippines. *The World Bank Economic Review*, *21*(2), 219–248.